

STAT 412 QZ07 — Question 8

Problem

A manufacturer claims that the average tensile strength of thread A exceeds the average tensile strength of thread B by at least 13 kilograms. Researchers wish to test this claim using a 0.01 level of significance. How large should the samples be if the power of the test is to be 0.9 when the true difference between thread types A and B is 6 kilograms? The population standard deviation for thread A is 6.35 kilograms and the population standard deviation for thread B is 5.59 kilograms.

Solution

This is a one-sided two-sample mean test with known population standard deviations. The distance between the claimed difference and the true difference used for power is

$$13 - 6 = 7.$$

For significance level $\alpha = 0.01$ and power 0.90,

$$z_{0.99} = 2.326, \quad z_{0.90} = 1.282.$$

For equal sample sizes,

$$n = \frac{(z_{0.99} + z_{0.90})^2(\sigma_A^2 + \sigma_B^2)}{(13 - 6)^2}.$$

Substitute the values:

$$n = \frac{(2.326 + 1.282)^2(6.35^2 + 5.59^2)}{7^2} = 19.01.$$

Round up:

$$n = 20.$$

Final Answer

The minimum sample size required is 20 for each sample.